

Stroke Prevention

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Stroke is one of the leading causes of death and long-term disability worldwide, with individuals aged 65 and older bearing a disproportionately high burden of disease. The risk of stroke more than doubles with each successive decade of life after age 55, and approximately 75% of all strokes occur in persons aged 65 or older. In the United States alone, nearly 800,000 strokes occur each year, of which an estimated 610,000 are first-time events.

The consequences of stroke in older adults extend far beyond the acute event. Survivors frequently face significant morbidity, including motor impairment, aphasia, cognitive decline, depression, and loss of independence — outcomes that impose immense personal, familial, and socioeconomic costs. Importantly, however, epidemiological research consistently demonstrates that 80% or more of strokes are preventable through evidence-based lifestyle modification and appropriate medical management.

Stroke: Types and Mechanisms

Accounting for approximately 87% of all strokes, ischemic strokes result from the obstruction of cerebral blood flow, typically due to:

- Cardioembolic mechanisms — thrombi originating from the heart (often associated with atrial fibrillation) that embolize to cerebral vessels.
- Large-artery atherosclerosis — plaque formation and stenosis in carotid or intracranial arteries.
- Small-vessel (lacunar) disease — occlusion of small penetrating arteries, commonly associated with chronic hypertension and diabetes.
- Cryptogenic stroke — ischemic events of undetermined etiology, increasingly recognized in older adults.

Hemorrhagic strokes (approximately 13% of cases) include:

- Intracerebral hemorrhage (ICH) — bleeding directly into brain tissue, strongly associated with hypertension and anticoagulant use.
- Subarachnoid hemorrhage (SAH) — bleeding into the space surrounding the brain, often from a ruptured aneurysm.

A transient ischemic attack — commonly called a "mini-stroke" — involves a temporary interruption of cerebral blood flow producing stroke-like symptoms that resolve within 24 hours (typically within minutes). TIAs are medical emergencies: the ABCD² score and subsequent imaging are used to stratify short-term stroke risk. Approximately 10–15% of TIA patients experience a stroke within 90 days, with the highest risk in the first 48 hours.

Risk Factors

Hypertension is the single most important modifiable risk factor for stroke, responsible for approximately 54% of stroke deaths globally. In elderly adults, isolated systolic hypertension (systolic ≥ 140 mmHg with normal diastolic) is particularly prevalent and clinically significant.

Target blood pressure: Current guidelines (AHA/ACC 2017) recommend a target of $<130/80$ mmHg for most adults with established cardiovascular risk, including most older adults. However, individualized

targets are appropriate — for frail older adults or those with significant comorbidities, less aggressive targets (e.g., <140/90 mmHg) may be preferable to avoid hypotension and falls.

Management strategies:

- Regular home blood pressure monitoring
- Adherence to antihypertensive medications
- Sodium restriction
- DASH dietary pattern
- Weight management
- Aerobic exercise
- Limitation of alcohol consumption

Atrial fibrillation is the most common sustained cardiac arrhythmia in older adults, affecting approximately 10% of individuals over age 80. AF increases stroke risk five-fold and is associated with more severe and disabling strokes than other etiologies.

Prevention and management:

- Systematic screening for AF using pulse palpation, ECG, or wearable devices
- Oral anticoagulation therapy — warfarin or direct oral anticoagulants (DOACs) — in appropriate candidates, guided by the CHA₂DS₂-VASc score
- Rate or rhythm control strategies as indicated
- Regular cardiology follow-up

Elevated low-density lipoprotein (LDL) cholesterol promotes atherogenesis in carotid and cerebral arteries. In older adults with prior stroke, TIA, or significant atherosclerotic cardiovascular disease (ASCVD), statin therapy is a cornerstone of secondary prevention.

Recommendations:

- Fasting lipid panel assessment
- Statin therapy for high-risk individuals (ASCVD risk ≥7.5% over 10 years)
- Dietary modification: reduce saturated and trans fats; increase omega-3 fatty acids and plant sterols

Type 2 diabetes confers a two- to four-fold increase in stroke risk through mechanisms including accelerated atherosclerosis, endothelial dysfunction, and hypercoagulability.

Management targets:

- HbA1c goal individualized (generally 7–7.5% in elderly; less stringent in frail patients to avoid hypoglycemia)
- Blood pressure and lipid control are particularly important in diabetic patients
- Regular monitoring for diabetic complications including renal disease

Central adiposity, insulin resistance, dyslipidemia, and hypertension collectively constitute metabolic syndrome, a recognized cluster of stroke risk factors.

Targets:

- Body mass index (BMI): 18.5–24.9 kg/m² (note: in adults >70, modest overweight [BMI 25–30] may not confer excess mortality risk)
- Waist circumference: <88 cm in women; <102 cm in men
- Emphasis on visceral fat reduction through dietary changes and exercise

Significant stenosis (≥50%) of the internal carotid artery is an important cause of ischemic stroke. Asymptomatic carotid stenosis is prevalent in older adults.

Management:

- Carotid duplex ultrasonography for at-risk individuals
- Optimal medical therapy (antiplatelet agents, statins, antihypertensives)
- Carotid endarterectomy or stenting in selected symptomatic patients with high-grade stenosis (≥70%)

Cigarette smoking doubles the risk of ischemic stroke. The risk is dose-dependent and persists with passive (secondhand) smoke exposure. Critically, smoking cessation substantially reduces stroke risk within 2–5 years, and benefits are observed at any age.

Cessation strategies:

- Nicotine replacement therapy (patches, gum, lozenges)
- Prescription pharmacotherapy (varenicline, bupropion)
- Behavioral counseling and support programs
- Motivational interviewing

The relationship between alcohol and stroke is complex. Heavy alcohol consumption (>2 standard drinks/day) is clearly associated with elevated stroke risk — both ischemic (via hypertension and AF) and hemorrhagic. Light-to-moderate intake may have neutral or modest cardiovascular effects, though current evidence does not support recommending alcohol consumption for prevention.

Recommendation: Limit alcohol to ≤1 drink/day for women and ≤2 drinks/day for men; abstinence is preferable for those with hypertension, AF, or liver disease.

Obstructive Sleep Apnea, characterized by recurrent nocturnal hypoxia and sleep fragmentation, is strongly associated with hypertension, cardiac arrhythmias, and stroke. OSA is markedly underdiagnosed in older adults.

Management:

- Screening using validated tools (STOP-BANG questionnaire)
- Polysomnography for diagnosis
- Continuous positive airway pressure (CPAP) therapy
- Weight loss and positional therapy as adjuncts

Dietary Strategies for Stroke Prevention

Diet plays a central role in modulating nearly every vascular risk factor relevant to stroke prevention. The following evidence-based dietary patterns and specific recommendations are particularly applicable to elderly adults.

The Dietary Approaches to Stop Hypertension (DASH) diet is one of the most rigorously studied dietary interventions for cardiovascular risk reduction. It is characterized by:

- High intake of fruits and vegetables (8–10 servings/day)
- Emphasis on whole grains (6–8 servings/day)
- Inclusion of low-fat dairy products (2–3 servings/day)
- Lean proteins — poultry, fish, legumes
- Nuts and seeds (4–5 servings/week)
- Strict sodium limitation ($\leq 1,500$ – $2,300$ mg/day)
- Limited red meat, sweets, and sugar-sweetened beverages

Clinical trials have demonstrated that the DASH diet reduces systolic blood pressure by 8–14 mmHg, with effects comparable to single-agent antihypertensive therapy.

The Mediterranean dietary pattern is associated with a 30% reduction in major cardiovascular events, including stroke, compared to low-fat diets (PREDIMED trial). Key features include:

- Abundant olive oil as the primary fat source (extra virgin preferred)
- High consumption of vegetables, fruits, legumes, and whole grains
- Regular intake of fish and seafood (≥ 2 servings/week)
- Moderate consumption of red wine with meals (optional; benefits do not justify initiating alcohol use)
- Low intake of red and processed meats
- Moderate dairy consumption (primarily yogurt and cheese)

The anti-inflammatory and antioxidant properties of the Mediterranean diet appear to reduce atherogenesis, improve endothelial function, and lower thrombotic risk.

Excess dietary sodium is the primary dietary driver of hypertension. The average American consumes approximately 3,400 mg/day — more than double the recommended maximum for high-risk individuals.

Practical reduction strategies:

- Cook from whole foods rather than processed or packaged products
- Read food labels; choose items with < 140 mg sodium per serving
- Use herbs, spices, citrus, and vinegar as flavor alternatives
- Limit restaurant meals, fast food, canned soups, and cured meats
- Use potassium-enriched salt substitutes (with caution in patients with renal impairment or on ACE inhibitors/ARBs)

Nutrient/Food	Evidence for Stroke Prevention	Dietary Sources
Omega-3 fatty acids	Reduces triglycerides, platelet aggregation, and inflammation	Fatty fish (salmon, mackerel, sardines), walnuts, flaxseed
Potassium	Lowers blood pressure; may directly reduce stroke risk	Bananas, avocado, sweet potato, leafy greens
Magnesium	Associated with reduced stroke risk; supports vascular function	Nuts, seeds, whole grains, dark leafy vegetables
Dietary fiber	Reduces LDL cholesterol and hypertension	Oats, beans, lentils, vegetables, fruits
Flavonoids/polyphenols	Antioxidant and anti-inflammatory effects	Berries, dark chocolate ($\geq 70\%$), tea, citrus
Folate and B vitamins	Reduces homocysteine (a vascular risk marker)	Leafy greens, legumes, fortified cereals
Calcium & Vitamin D	Supports vascular health; reduces fracture risk	Dairy, fortified plant milks, fatty fish, sun exposure

Foods and Substances to Limit or Avoid

- Saturated and trans fats — promote atherogenesis; found in red meat, full-fat dairy, processed foods, and fried foods
- Added sugars — contribute to obesity, metabolic syndrome, and inflammation
- Ultra-processed foods — associated with elevated stroke risk independent of individual nutrients
- Excessive caffeine — may transiently raise blood pressure; moderate intake (≤ 3 cups/day) is generally acceptable
- High-sodium condiments — soy sauce, ketchup, pickled foods, canned vegetables

Hydration

Dehydration in older adults contributes to hemoconcentration and increased blood viscosity, elevating thrombotic risk. Older adults have reduced thirst sensation and are particularly vulnerable.

Recommendation: Aim for 6–8 cups (1.5–2 liters) of fluid per day, primarily water; adjust for climate, activity level, and medical conditions (e.g., heart failure, renal disease).

Physical Activity

Regular physical activity is one of the most potent interventions for stroke prevention, with benefits spanning virtually every modifiable risk factor. Sedentary behavior is an independent stroke risk factor, and even modest increases in physical activity confer meaningful protection.

Benefits of Exercise in Older Adults

- Reduces systolic blood pressure by 4–9 mmHg
- Improves lipid profiles (raises HDL, lowers LDL and triglycerides)
- Enhances insulin sensitivity and glycemic control
- Facilitates weight management

- Reduces inflammation (lower CRP and inflammatory cytokines)
- Improves cardiac function and reduces AF risk
- Enhances endothelial function and arterial compliance
- Reduces stress, anxiety, and depression

Aerobic (Cardiovascular) Exercise

- Minimum target: 150 minutes per week of moderate-intensity aerobic activity, or 75 minutes per week of vigorous-intensity aerobic activity, or an equivalent combination
- Ideal target: 300 minutes per week of moderate-intensity activity for additional health benefits
- Examples of moderate-intensity activities: Brisk walking, water aerobics, recreational cycling, dancing, gardening
- Examples of vigorous-intensity activities: Jogging, swimming laps, aerobics classes, cycling with hills

For sedentary older adults, even **10-minute bouts** of walking distributed throughout the day confer cardiovascular benefit and represent an appropriate starting point.

Resistance (Strength) Training

- Frequency: At least 2 days per week, targeting all major muscle groups
- Benefits: Improves metabolic health, preserves lean muscle mass (sarcopenia prevention), supports functional independence, and reduces falls risk
- Examples: Body-weight exercises (sit-to-stand, wall push-ups), resistance bands, light free weights, weight machines
- Precaution: Avoid the Valsalva maneuver (breath-holding during exertion), which transiently elevates blood pressure

Flexibility and Balance Training

- Frequency: Daily or most days of the week
- Benefits: Reduces fall risk (and consequent head trauma), maintains range of motion, and supports functional mobility
- Examples: Static stretching, yoga, tai chi, Pilates
- Tai Chi: Deserves special mention — multiple randomized controlled trials demonstrate that tai chi significantly reduces falls, improves balance, lowers blood pressure, and reduces psychological stress in older adults

Sedentary Behavior Reduction

Beyond structured exercise, limiting prolonged sedentary time is independently important. Research suggests that sitting for more than 8 hours per day is associated with elevated cardiovascular risk, even among individuals who exercise regularly.

Practical strategies:

- Set an alarm to stand and move for 2–3 minutes every 30–45 minutes
- Use a standing desk when possible
- Walk during phone calls
- Replace television time with light activity (gentle stretching, walking in place)

Healthy Lifestyle Habits

Smoking Cessation

Smoking cessation at any age is beneficial. Within 2 years of quitting, stroke risk decreases substantially; by 5 years, risk approaches that of a non-smoker. Healthcare providers should employ the 5 A's framework (Ask, Advise, Assess, Assist, Arrange) at every clinical encounter.

Sleep Hygiene

Quality sleep is essential for cardiovascular health. Both insufficient sleep (<6 hours) and excessive sleep (>9 hours) have been associated with elevated stroke risk. Obstructive sleep apnea — extremely prevalent in older adults — should be actively identified and treated (see Section 3.2.9).

Sleep hygiene recommendations:

- Maintain consistent sleep and wake times
- Keep the bedroom cool, dark, and quiet
- Avoid caffeine after noon and alcohol close to bedtime
- Limit screen exposure within 1 hour of sleep
- Aim for 7–8 hours of sleep per night

Stress Management

Chronic psychological stress activates the hypothalamic-pituitary-adrenal (HPA) axis, elevates cortisol, promotes inflammation, raises blood pressure, and increases thrombotic risk. Acute stress events may precipitate stroke.

Evidence-based stress reduction strategies:

- Mindfulness-Based Stress Reduction (MBSR): Demonstrated to lower blood pressure and reduce anxiety in older adults
- Diaphragmatic breathing: Activates parasympathetic nervous system; reduces acute stress response
- Progressive muscle relaxation
- Social engagement and meaningful activity
- Journaling and cognitive reframing

- Professional counseling for persistent anxiety or depression

Medication Adherence

Non-adherence to prescribed medications — antihypertensives, anticoagulants, antiplatelets, and statins — is among the most common and preventable causes of stroke recurrence. Older adults face particular challenges due to polypharmacy, cognitive changes, cost, and pill burden.

Strategies to improve adherence:

- Pill organizers and blister packs
- Electronic medication reminder alarms or apps
- Simplification of medication regimens (once-daily dosing where possible)
- Regular pharmacist medication reviews
- Caregiver involvement and supervision

Fall Prevention

Falls are a leading cause of injury and mortality in older adults and are particularly hazardous for individuals on anticoagulant therapy, where intracranial hemorrhage is a feared complication. Fall prevention is therefore an integral component of stroke-related care.

Evidence-based fall prevention:

- Exercise programs targeting balance and strength (tai chi, structured resistance training)
- Home hazard assessment and modification — removal of trip hazards, installation of grab bars, improved lighting
- Footwear optimization — supportive, low-heeled shoes with non-slip soles
- Vision and hearing correction
- Medication review — identify and deprescribe fall-risk medications (benzodiazepines, sedatives, anticholinergics, diuretics)
- Vitamin D supplementation — emerging evidence supports vitamin D in reducing fall risk (discuss with physician)

Alcohol Moderation

As detailed in Section 3.2.8, alcohol consumption should be moderate at most. In patients with AF, liver disease, or prior hemorrhagic stroke, abstinence is recommended. Older adults metabolize alcohol more slowly and are more susceptible to its adverse effects (falls, drug interactions, cognitive impairment) than younger adults.